

# Mass, Energy, and Nuclear Reactions

NE 630

## Lesson 01

**Prepare:** Read FNRP Sections 1.1–1.3. Define the key terms in your notes and use the learning objectives to guide the reading.

### Macro objective

Determine and quantify the reactants, products, and energy inputs/outputs of nuclear reactions.

**Vocabulary:** nuclide; atomic number; atomic mass; mass number; nucleon; rest mass; total energy; kinetic energy; binary reaction; radioactive decay;  $Q$  value; binding energy; energy density

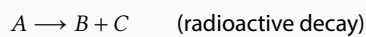
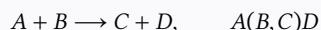
### 1 Nuclides and reaction language

#### Read the symbol

$${}^A_ZX, \quad A = Z + N$$

$X$  .....  
 $Z$  .....  
 $A$  .....  
 $N$  .....

#### Reaction forms used in this lesson



Organize the initial and final values before using mass data:

|               | initial | final |
|---------------|---------|-------|
| mass number   | .....   | ..... |
| atomic number | .....   | ..... |

### 2 Mass change and $Q$ values

#### Relations and conversion

$$\Delta E = \Delta mc^2, \quad Q = \left( \sum m_{\text{initial}} - \sum m_{\text{final}} \right) c^2$$
$$1 \text{ u} = 931.5 \text{ MeV}/c^2$$

The reaction rearranges bound nucleons (or electrons in a chemical reaction). Nuclear changes are typically on the MeV scale; chemical changes are typically on the eV scale. **Board check:** If  $Q > 0$ , which mass sum is larger, and what does that imply for a decay?

### 3 Relating neutron energies

#### Exact and classical expressions

$$E = mc^2, \quad m = \frac{m_0}{\sqrt{1 - (v/c)^2}}, \quad \beta = \frac{v}{c}$$

$$K_{\text{true}} = m_0 c^2 \left( \frac{1}{\sqrt{1 - \beta^2}} - 1 \right), \quad K_{\text{classical}} = \frac{1}{2} m_0 v^2$$

$$\% \text{ err} = 100 \frac{K_{\text{classical}} - K_{\text{true}}}{K_{\text{true}}}$$

**Neutron data:**  $m_0 = 1.008664 \text{ u}$ ,  $m_0 c^2 \approx 939.57 \text{ MeV}$ ,  $c = 2.9979 \times 10^8 \text{ m/s}$ .

#### Speed associated with a given $K_{\text{true}}$

Starting from the exact kinetic energy, isolate  $\beta^2$ :

$$\beta^2 = 1 - \left( \frac{m_0 c^2}{K_{\text{true}} + m_0 c^2} \right)^2 = \frac{v^2}{c^2}.$$

Use this result to express  $K_{\text{classical}}$  entirely in terms of  $K_{\text{true}}$  and  $m_0 c^2$ .

#### Threshold question

Find the maximum true neutron kinetic energy for which the magnitude of the classical error is no more than 1%.

**Board result:**  $K_{\text{true}} \approx \dots\dots\dots$

#### Homework bridge: a 2.0 MeV fission neutron

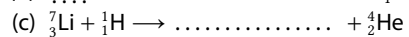
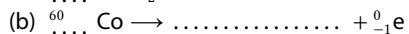
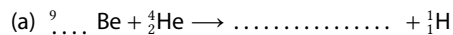
|                        |       |       |       |
|------------------------|-------|-------|-------|
| $\beta^2$              | ..... | $v$   | ..... |
| $K_{\text{classical}}$ | ..... | error | ..... |

#### Interpret the approximation

The classical expression ..... the exact kinetic energy. Connect the 1% threshold to the lesson objective that neutron kinematics below about 10 MeV can often be treated classically.

4 Reaction completion and  $Q$ -value practiceComplete each reaction; then compute  $Q$ 

Use one reaction as the board example; retain the others for practice.



$Q_a$  .....  $Q_b$  .....

$Q_c$  .....  $Q_d$  .....

## Atomic-mass data used in the assigned problems (u)

| nuclide            | mass           | nuclide            | mass          |
|--------------------|----------------|--------------------|---------------|
| ${}^1\text{H}$     | 1.007825031898 | ${}^4\text{He}$    | 4.00260325413 |
| ${}^7\text{Li}$    | 7.01600343426  | ${}^9\text{Be}$    | 9.012183062   |
| ${}^{10}\text{B}$  | 10.012936862   | ${}^{12}\text{B}$  | 12.014352638  |
| ${}^{13}\text{C}$  | 13.00335483534 | ${}^{60}\text{Co}$ | 59.933815536  |
| ${}^{60}\text{Ni}$ | 59.930785129   | $n$                | 1.008664      |

## Atomic masses and decay

Before evaluating the  $\beta^-$ -decay  $Q$  value with the supplied *atomic* masses, show which electron masses are already contained in the parent and daughter entries.

## Mass defect and binding energy per nucleon

Reconstruct the definitions from Sections 1.1–1.3:

$\Delta m =$  .....

$B =$  .....

$B/A =$  .....

Use  ${}^4_2\text{He}$  and the fixed mass data as a quick check.

## 5 Energy density of reactions

## Plant-scale fixed inputs from the board notes

|                | nuclear plant                  | coal plant        |
|----------------|--------------------------------|-------------------|
| electric power | 1 GW <sub>e</sub>              | 1 GW <sub>e</sub> |
| efficiency     | 0.30                           | 0.40              |
| material basis | 100 t UO <sub>2</sub> over 5 y | 10,000 t per day  |

## Energy per unit material mass

Use  $E = P \Delta t$  and account for thermal efficiency:

$$\left. \frac{E}{m} \right|_{\text{nuc}} \approx \frac{1 \times 10^9 \text{ W}_e}{0.30} \frac{5(365)(24) \text{ h}}{100 \times 10^3 \text{ kg}} = \dots$$

$$\left. \frac{E}{m} \right|_{\text{coal}} \approx \frac{1 \times 10^9 \text{ W}_e}{0.40} \frac{24 \text{ h}}{10,000 \times 10^3 \text{ kg}} = \dots$$

Ratio and interpretation: .....

## Reaction-scale mass conversion

$$\left. \frac{\Delta m}{m_0} \right|_{\text{fission}} \approx \frac{200}{236(931.5)} = \dots$$

$$\left. \frac{\Delta m}{m_0} \right|_{\text{combustion}} \approx \frac{4 \times 10^{-6}}{44(931.5)} = \dots$$

How do these fractions explain the plant-scale comparison?

## Three related scales

For pure matter,

$$\frac{E}{m} = c^2 \approx \dots$$

From the board notes,

$$\left. \frac{P}{m} \right|_{\text{nuc}} \approx \dots, \quad \left. \frac{P}{m} \right|_{\text{coal}} \approx \dots$$

Why is the power-density contrast smaller than the energy-density contrast?